

Does Vowel Diacritization Hurt LLM Question Answering? Evidence, Failure Modes, and Metric Artifacts in Hebrew and Arabic

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Abstract

Hebrew and Arabic are typically written without vowel diacritics (Niqud / Tashkeel), which appear only in formal, liturgical, or disambiguation-critical text. We ask whether adding diacritics to QA inputs helps or hurts LLMs, testing three models across six benchmarks in both languages, with datasets vocalized using SOTA automatic vocalizers. Our results show that diacritization consistently degrades Arabic QA accuracy, while in Hebrew, the raw signal looked noisy: Niqud produced a positive delta on one dataset but a negative delta on the others. Re-scoring with an LLM-as-judge sharpened the signal—and closer inspection revealed that the effect is not uniform even in Arabic, but rather a mix of model-specific causes. An ablation that moves each word’s diacritics after its consonants, while holding their content fixed, substantially reduces both this collapse and the accuracy drop—evidence that how diacritics interrupt word forms, rather than their mere presence, contributes to the damage. Throughout the paper, we work toward classifying the diacritic-induced failure modes we observe across models. We release a corpus of judge-confirmed diacritic-induced failure cases for future mechanistic analysis.

1 Introduction

Hebrew and Arabic are usually written without the small marks that spell out vowels (Niqud in Hebrew, Tashkeel in Arabic); readers infer the vowels from context. These marks appear precisely where correct interpretation matters most: children’s books, religious texts, poetry, and other disambiguation-critical writing. We ask whether adding these vowel marks to a passage, before it is given to a language model, changes the model’s ability to answer questions about that passage.

To our knowledge, no prior published work directly studies the effect of vowel diacritization on LLMs for Hebrew and Arabic QA. At first glance, we found only a weak correlation between model

failure on vocalized QA and subword fragmentation ($r = 0.13$), based on dataset-level F1/Exact Match (EM) scores. These results suggested that the effect of diacritization cannot be explained by subword fragmentation alone, motivating a more fine-grained analysis of its impact on model behavior.

We revisit the question at the per-example level with three additions: **a postfix ablation that moves each word’s diacritics after its consonants** while holding their content fixed, testing whether the way diacritics interrupt word forms, rather than their mere presence, contributes to the degradation; **an LLM-as-judge re-scoring pass** that tests whether F1/EM’s conclusions hold under a semantic, rather than lexical, correctness check; and **manual inspection** of individual failure and success cases across both languages. It turns out that vocalization does not produce one uniform effect but a range of model-specific failure modes — and occasional genuine improvements — which we try to classify, e.g. a fixed stock-phrase collapse resembling sentiment classification (SILMA, Arabic), a shift toward code-switching into Chinese (Qwen, Hebrew), and — surprisingly — shorter, more accurate answers (dictalm2, Hebrew). The LLM-as-judge does not replace this classification effort; it sharpens it by giving us a more reliable correctness signal to classify against.

Any system serving Hebrew or Arabic readers of diacritized text — religious, educational, or literary content among them — risks a real accuracy degradation that standard benchmark scores can obscure or even contradict.

Our contributions: (i) we classify the model-specific failure — and occasional success — modes that diacritization induces across Hebrew and Arabic, including collapse into fixed, task-irrelevant stock answers, cross-lingual code-switching, and unexpected answer-shortening effects; (ii) we show that this failure is not explained by the mere presence of diacritics: an ablation that holds diacritic content fixed and moves each

Code, data, and more can be found on our GitHub:
<https://github.com/ElyasNM/diacritics-llm-ANLP>

word’s marks after its consonants more than halves the collapse rate and reduces the accuracy drop by roughly a third to a half, though the precise mechanism behind this effect remains an open question; (iii) we mine and release ~ 900 judge-confirmed diacritic-induced failure cases as a resource for follow-up interpretability work (logit-lens / patchscopes probing of entity retrieval). As part of the study, we also diacritized all six evaluated datasets and release the resulting versions alongside our code, enabling reproduction and further investigation of the observed effects.

2 Data

We evaluate on three QA benchmarks per language, all drawn from existing, publicly available, human-authored benchmarks rather than data we collected or generated ourselves. **Arabic:** ARCD (Mozannar et al., 2019) ($n=1,395$), TyDiQA-Arabic (Clark et al., 2020) ($n=15,726$), and MKQA-Arabic (Longpre et al., 2021). **Hebrew:** HeQ (Cohen et al., 2023) ($n=8,363$), ParaShoot (Keren and Levy, 2021) ($n=3,038$), and MKQA-Hebrew (Longpre et al., 2021) ($n=4,573$). Together, these benchmarks span extractive QA and reading comprehension, with open-domain QA.

For each dataset we construct a paired evaluation: the identical example is presented to the model with and without diacritics, so that any accuracy difference is attributable to the diacritics alone. Diacritics were added automatically, not by human annotators: Hebrew Niqud was applied with DICTA’s dictabert-large-charmenaked (Shmidman et al., 2020), and Arabic Tashkeel with Farasa (Abdelali et al., 2016). Table 1 reports per-dataset sizes.

Dataset	Language	n
ARCD	Arabic	1,395
TyDiQA	Arabic	15,726
MKQA-AR	Arabic	4,966
HeQ	Hebrew	8,363
ParaShoot	Hebrew	3,038
MKQA-HE	Hebrew	4,573

Table 1: Per-dataset example counts.

3 Methods

Models. Arabic: SILMA-9B-Instruct (SILMA.AI, 2024) (a model natively fine-tuned for Arabic), Qwen2.5-7B-Instruct (Team, 2024), and dictalm2.0-instruct (Shmidman et al., 2024)

(a model fine-tuned for Hebrew, used here as a cross-lingual control on Arabic). Hebrew: Qwen2.5-7B-Instruct (Team, 2024) and dictalm2.0-instruct (Shmidman et al., 2024) (fine-tuned natively for Hebrew).

Primary metric. Token-level F1 and exact match (EM) between the model’s generated answer and the gold span, computed separately for the plain and diacritized condition and reported as $\Delta = \text{diacritized} - \text{plain}$.

Fragmentation ratio. For each model/dataset we compute the ratio of subword-tokenizer token count on diacritized vs. plain text, as a proxy for how much diacritics inflate the input’s effective length.

Postfix ablation. In standard Tashkeel, each diacritic sits directly after the consonant it modifies, interleaved with the letters of the word. We generate an alternative encoding that moves all diacritics of each word to follow its consonant sequence, preserving which marks appear in each word while changing how they interrupt the consonant skeleton, and re-run SILMA on the same examples.

Collapse mining. We frequency-rank each model’s diacritized-condition predictions to detect degenerate, task-irrelevant stock answers repeated far more often than plain-condition predictions.

LLM-as-judge. Because F1/EM penalize any lexical deviation from the gold span even when an answer is semantically correct (or credit partial n-gram overlap even when an answer is semantically wrong), we additionally re-score a stratified sample (500–600 examples per dataset/model, matched pairs across the plain/diacritized conditions) with gpt-4o-mini-2024-07-18 (OpenAI), accessed via the OpenAI API, acting as a binary CORRECT/INCORRECT judge given the question, gold answer(s), and model prediction; the same judge model and prompt were used for both the Hebrew and Arabic samples. We report judge accuracy and Δ alongside F1/EM. Judge reliability was checked against human labels (Section 5).

4 Results

4.1 Arabic results

Under standard (prefix) Tashkeel, every model degrades on every dataset (Table 2); dictalm2 — a Hebrew-tuned model with no native Arabic pre-training, used here as a cross-lingual control — takes the largest F1 hit despite having the smallest

tokenizer fragmentation of the three (Table 4).

4.2 Hebrew results

Under Niqud, the raw picture is mixed rather than uniform: both models gain on HeQ but lose on ParaShoot and MKQA (Table 3) — the source of the “no effect” reading the pooled aggregate invites, examined in Section 4.5.

Model	Ds	EM _P	EM _N	ΔEM	F1 _P	F1 _N	ΔF1
Qwen	HeQ	10.09	11.99	+1.90	16.68	20.35	+3.68
Qwen	PS	27.49	19.32	-8.16	58.04	44.88	-13.15
Qwen	MKQA	1.70	0.94	-0.77	5.37	3.34	-2.03
Qwen	All	11.00	10.22	-0.78	21.31	20.15	-1.16
Dicta	HeQ	7.29	11.97	+4.68	22.64	27.10	+4.46
Dicta	PS	10.63	10.37	-0.26	47.63	41.54	-6.09
Dicta	MKQA	8.31	1.90	-6.41	16.90	7.39	-9.51
Dicta	All	8.22	8.78	+0.56	25.75	24.20	-1.55

Table 3: Hebrew: EM/F1, plain vs. Niqud, per model and dataset ($n=15,974$ total). PS = ParaShoot.

4.3 Why the raw signal diverges

Arabic Tashkeel degrades every model on every dataset; Hebrew Niqud is noisier, helping on HeQ while hurting ParaShoot and MKQA. Fragmentation ratio does not explain either pattern: Table 4 shows the *lowest*-fragmentation Arabic model (dictalm2) taking the *largest* hit, and §4.5 shows Hebrew and Arabic converge to a similar magnitude once judge-scored, despite Hebrew’s far higher fragmentation ratios for the same architectures — fragmentation ratio is not the driver in either language.

Model (Arabic)	Frag. ratio	ΔF1
dictalm2	1.53–1.55×	-6.48
SILMA-9B	2.54–2.62×	-5.91
Qwen2.5-7B	2.67–2.81×	-4.95

Table 4: Fragmentation ratio (plain→Tashkeel token inflation) vs. overall ΔF1, ranked ascending. Pearson $r=0.13$ across all model×dataset pairs.

4.4 Failure Modes, Recovery Cases, and Deeper Inspection of Diacritization Effects

SILMA: sentiment-phrase collapse (Arabic). Under Tashkeel, SILMA collapses onto a fixed stock phrase on 5.5% of ARCD+TyDiQA questions (0.5% under plain text, $\sim 11\times$), answering unrelated factual questions as if performing sentiment classification rather than QA:

Q: ما هي الهندسة الميكانيكية ؟ ("What is mechanical engineering?") — **Gold:** a branch of engineering concerned with design,

manufacture, and operation of machines — **Pred:** هو رأي سلبي ("it is a negative opinion")

Q: من هم الرونن ؟ ("Who are the ronin?") — **Gold:** a samurai without a lord — **Pred:** هو رأي سلبي ("it is a negative opinion")

Postfix re-encoding (Table 5), which keeps the same diacritic marks in each word but moves them out of the consonant skeleton, more than halves the collapse rate (5.43% → 2.29%) and reduces ΔF1 by $\sim 40\%$ ($-6.39 \rightarrow -3.84$) and ΔEM by $\sim 31\%$ ($-8.04 \rightarrow -5.57$), suggesting that how diacritics interrupt word forms contributes to SILMA’s degradation.

SILMA (Arabic)	Prefix	Postfix
ΔF1	-6.39	-3.84
ΔEM	-8.04	-5.57
Collapse rate	5.43%	2.29%
Frag. ratio	2.54–2.62×	~ 13 –14% lower

Table 5: Moving each word’s diacritics after its consonants (content unchanged) vs. SILMA’s Arabic degradation and collapse rate (plain-baseline collapse rate $\approx 0.5\%$ in both).

Qwen: code-switch collapse (Hebrew, HeQ).

On HeQ, 23.3% (140/600) of Qwen’s *plain*-condition predictions degrade into a Chinese-language refusal — more than double the 10.8% (65/600) rate under Niqud — and not one of the 140 is ever judged correct:

Q: GETT? באיזו שנה הוקמה ("In what year was GETT founded?") — **Gold:** 2013
Plain: הוקמה GETT אך הקטעעפס safeged המ
给你们提供的文本并没有提到GETT的成立年份...
("...the passage does not state GETT’s founding year...") Niqud: 2013 (correct)

Q: מה היא החברה הרווחית ביותר בכל הזמנים? ("What is the most profitable company of all time?") — **Gold:** אפל (Apple)
Plain: התשובה אינה מופיעה
所提供的段落并没有提到全球最赚钱的公司相关信息...
("...the passage does not mention the world’s most profitable company...") Niqud: אפל ("Apple," correct)

This runs *opposite* to SILMA’s collapse — it fires *without* diacritics — and is HeQ-specific (ParaShoot: 14/500 vs. 12/500; MKQA: 19/400 vs. 29/400, both small and symmetric). It is also the full explanation for Qwen’s apparent Niqud “help” on HeQ (§4.5).

dictalm2: shorter, more accurate answers (Hebrew, HeQ). Under Niqud, dictalm2’s HeQ answers get shorter and land on the gold span more

Model	Dataset	EM _{plain}	EM _{Tash}	Δ EM	F1 _{plain}	F1 _{Tash}	Δ F1
Qwen2.5-7B	ARCD	27.46	21.36	-6.09	64.66	58.90	-5.76
Qwen2.5-7B	TyDiQA	22.73	18.85	-3.88	56.96	50.80	-6.16
Qwen2.5-7B	MKQA	2.15	1.55	-0.60	6.97	6.11	-0.87
Qwen2.5-7B	Overall	18.40	15.12	-3.28	46.21	41.26	-4.95
SILMA-9B	ARCD	41.86	37.28	-4.59	70.52	64.01	-6.51
SILMA-9B	TyDiQA	48.78	40.55	-8.23	68.06	61.69	-6.37
SILMA-9B	MKQA	6.56	4.33	-2.24	12.12	7.83	-4.29
SILMA-9B	Overall	38.85	32.20	-6.65	55.64	49.73	-5.91
dictalm2	ARCD	11.83	8.89	-2.94	48.46	39.96	-8.50
dictalm2	TyDiQA	16.49	13.41	-3.08	48.34	40.19	-8.15
dictalm2	MKQA	1.15	0.50	-0.64	2.90	2.28	-0.62
dictalm2	Overall	12.75	10.22	-2.53	38.13	31.65	-6.48

Table 2: Arabic: EM/F1, plain vs. Tashkeel, per model and dataset ($n=22,087$ total). Every Δ is negative.

often; EM rewards only exact spans, so a ~~terse~~ register lifts EM whenever the plain-condition answer wraps the identical fact in a full sentence:

Q: מתי החלה תנועת המחאה האפרו אמריקאית? ("When did the Afro-American protest movement begin?") — **Gold:** 1965
Niqud: 1965 ($EM=1$) Plain: תנועת המחאה
.1965 האפרו-אמריקאית החלה ב- ("...began in 1965." $EM=0$ — right content, wrong form)

Q: כמה עלינו לשתות ביום? ("How much should we drink per day?") — **Gold:** 4 בקבוקי שתייה
(4 bottles)
Niqud: 4 בקבוקי שתייה ($EM=1$) Plain: אנחנו
צריכים לשתות לפחות 3 מחסניות של Cirkul ביום.
("...at least 3 Cirkul cartridges..." $EM=0$, wrong content)

We initially attributed the apparent HeQ gain to the shorter, more compact answers dictalm2 produces under Niqud (Section 4.4, "shorter, more accurate answers"): EM rewards a terse span that lands exactly on the gold answer, while the plain-condition answer often wraps the same fact in a full sentence and is penalized despite being equally correct. Re-scoring with the LLM-as-judge, however, reveals that this explanation only partly survives: for dictalm2, the judge-based gap on HeQ nearly flattens ($+4.46$ F1 $\rightarrow +1.17$ judge- Δ), confirming that most of the raw gain was a surface-form artifact of EM/F1 rewarding brevity rather than genuine comprehension. For Qwen, the picture is different again: its HeQ "gain" turns out to be almost entirely a byproduct of the code-switch-into-Chinese collapse (Section 4.4) — a failure mode that, for reasons that remain unclear, fires less often once the input is vocalized.

Association-driven substitution. A word or proper noun inside the *vocalized* question itself pulls in an unrelated but strongly co-occurring entity, distinct from confusable substitution in that the wrong answer is traceable to a specific lexical

trigger in the prompt rather than mere domain proximity:

Q: מי היה הזמר המקורי של השיר הללויה? ("Who was the original singer of the song Hallelujah?") — **Gold:** Leonard Cohen —
Pred: דוד המלך ("King David" — pulled in via the biblical/Psalms association of the vocalized word הללויה)

Q: מי נלחם בקרב קדש? ("Who fought at the Battle of Kadesh?") — **Gold:** Ramesses II —
Pred: משה ("Moses" — pulled in via the Torah association of the vocalized place name קדש, rather than the historical battle it names)

General pattern: subject echo. Diacritics sometimes trigger the model to answer with the entity named in the question instead of retrieving the fact associated with it:

Q: איזו חברה משתמשת בשירותיה של BrandTotal? ("Which company uses BrandTotal's services?") — **Gold:** פייסבוק (Facebook)
Plain: פייסבוק (correct) Niqud: BrandTotal (echoes the queried entity itself)

Q: down a of system של מי הזמר של System of a Down? ("Who is the singer of System of a Down?") — **Gold:** סרג' טנקיאן (Serj Tankian)
Plain: סרג' טנקיאן (correct) Niqud: of System Down a ("System of a Down is the singer.")

More general patterns can be found in Appendix B.

Both are mined from the ~ 900 judge-confirmed harm pairs (§5) and motivate the logit-lens follow-up: does the correct entity remain linearly decodable before the model collapses onto the echo?

4.5 LLM-judge validation

Arabic. The judge (stratified sample, same three datasets) finds a uniformly larger effect than F1/EM but preserves the cross-model ranking (F1)

— dictalm2 > SILMA > Qwen (Table 6) — validating F1’s relative ordering even though it understates the absolute size.

Model	F1/EM Δ	Judge Δ
dictalm2	−6.48 / −2.53	−8.13
SILMA-9B	−5.91 / −6.65	−7.46
Qwen2.5-7B	−4.95 / −3.28	−6.27

Table 6: Arabic overall Δ (F1/EM vs. judge), ranked by judge magnitude.

Hebrew. The judge confirms F1/EM understated ParaShoot/MKQA harm (Table 7); its apparent HeQ ”help” is Qwen’s code-switch-collapse artifact (§4.4), not evidence Niqud aids comprehension; for dictalm2, the HeQ gain is likewise not significant under the judge (McNemar $p=0.625$). Excluding HeQ examples in which Qwen answers in Chinese under either condition, its HeQ effect falls from +3.5 to +0.7 pp ($n=424$, McNemar $p=0.84$). Hebrew’s overall judge-based effect (−5.67 Qwen, −8.93 dictalm2, combined $n=3,000$, $\Delta=-7.3$) is the same order of magnitude as Arabic’s.

Model	Ds	F1 Δ	Judge Δ
Qwen	HeQ	+3.68	+3.5
Qwen	PS	−13.15	−17.8
Qwen	MKQA	−2.03	−4.25
Qwen	All	−1.16	−5.67
dictalm2	HeQ	+4.46	+1.17
dictalm2	PS	−6.09	−16.6
dictalm2	MKQA	−9.51	−14.5
dictalm2	All	−1.55	−8.93

Table 7: Hebrew per-dataset Δ : F1 vs. judge. PS = ParaShoot.

5 Discussion, Limitations, and Future Work

The LLM-judge sample (500–600 examples/dataset/model) is smaller than the full F1/EM evaluation and was scored by a single judge model. We checked it against human labels in three validations: an Arabic binary pilot (94.4% agreement, Cohen’s $\kappa=0.874$, 15/15 consistency), a post-run sanity check on Arabic Qwen outputs (90.8%, $\kappa=0.813$), and a Hebrew pilot (94.0%, $\kappa=0.879$, 15/15 consistency). In every condition the judge saw the plain (unvocalized) question, regardless of the condition the model was run under. Even so, we treat its deltas as a validation signal for metric reliability rather than a replacement ground truth. A Qwen postfix-Tashkeel rerun — which would test whether the tokenization-placement effect generalizes beyond

SILMA — remains future work. We have mined ~ 900 judge-confirmed correct→incorrect pairs (427 Hebrew, 496 Arabic) and manually curated a subset with short, extractable entity answers (company/person/place names) as candidates for a logit-lens / patchscopes probe of whether the correct entity remains linearly decodable from intermediate hidden states even when the diacritized-condition output is wrong, motivated by the subject-echo pattern in Section 4.4. This mechanistic analysis is in progress and not yet part of this report’s results.

6 Conclusion

Vowel diacritization measurably degrades LLM QA accuracy in both Hebrew and Arabic, by a similar magnitude once evaluated with a metric robust to lexical-surface noise. In Arabic this degradation is metric-robust and partly explained by a model-specific stock-phrase collapse (SILMA) whose rate more than halves when each word’s diacritics are moved after its consonants, evidence that how diacritics interrupt word forms contributes to the degradation. In Hebrew, raw F1/EM aggregates are unreliable for two compounding reasons: they nearly cancel to zero due to opposing per-dataset effects and partial-credit scoring, and Qwen’s one apparently helpful dataset (HeQ) is itself a collapse artifact — a code-switch-into-Chinese failure mode that fires more often *without* diacritics, not evidence that diacritics help. Once both artifacts are accounted for, Hebrew shows a real, consistent degradation on the same order as Arabic’s; no significant effect of Niqud was found on HeQ once the Qwen collapse artifact is removed, and no significant effect was found for dictalm2 under the judge. Practically, this argues for (a) caution when using surface-overlap metrics to evaluate any morphologically rich, diacritic-sensitive language, (b) diacritic-aware input encoding or tokenization as a possible low-cost mitigation worth further study, and (c) checking apparent ”diacritics help” signals for collapse artifacts before trusting them.

References

- Ahmed Abdelali, Kareem Darwish, Nadir Durrani, and Hamdy Mubarak. 2016. [Farasa: A fast and furious segmenter for Arabic](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, pages 11–16, San Diego, California. Association for Computational Linguistics.
- Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Gar-

rette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. *Transactions of the Association for Computational Linguistics*, 8:454–470.

Amir DN Cohen, Hilla Merhav, Yoav Goldberg, and Reut Tsarfaty. 2023. HeQ: a large and diverse Hebrew reading comprehension benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13693–13705. Association for Computational Linguistics.

Omri Keren and Omer Levy. 2021. ParaShoot: A Hebrew question answering dataset. In *Proceedings of the 3rd Workshop on Machine Reading for Question Answering*, pages 106–112. Association for Computational Linguistics.

Shayne Longpre, Yi Lu, and Joachim Daiber. 2021. MKQA: A linguistically diverse benchmark for multilingual open domain question answering. *Transactions of the Association for Computational Linguistics*, 9:1389–1406.

Hussein Mozannar, Elie Maamary, Karl El Hajal, and Hazem Hajj. 2019. Neural Arabic question answering. In *Proceedings of the Fourth Arabic Natural Language Processing Workshop*, pages 108–118, Florence, Italy. Association for Computational Linguistics.

Avi Shmidman, Shaltiel Shmidman, Moshe Koppel, and Yoav Goldberg. 2020. Nakdan: Professional Hebrew diacritizer. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 197–203, Online. Association for Computational Linguistics.

Shaltiel Shmidman, Avi Shmidman, Amir DN Cohen, and Moshe Koppel. 2024. Adapting LLMs to Hebrew: Unveiling DictaLM 2.0 with enhanced vocabulary and instruction capabilities.

SILMA.AI. 2024. Silma 9b instruct v1.0. <https://huggingface.co/silma-ai/SILMA-9B-Instruct-v1.0>.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

A LLM-as-Judge Prompts

Both judges use gpt-4o-mini-2024-07-18, a temperature of 0, and a strict JSON output schema. The system prompt is written in English in both languages; only the worked examples and the language-specific spelling/diacritic exception differ. The Arabic and Hebrew judge runs are otherwise identical pipelines (blinded to model identity and experimental condition, no F1/EM signal passed in).

A.1 Hebrew judge

You are a Hebrew QA evaluation judge. Your task is to assess whether a candidate answer correctly answers a question, given the reference (gold) answer(s).

Judge the candidate based on whether it sufficiently and factually answers the question relative to the provided gold answer(s).

If a context passage is provided, use it as additional evidence. If no context is provided, judge based on the question and gold answer(s). Treat the provided gold answer(s) as the reference; do not override them using outside knowledge.

You must respond with valid JSON only. No explanation outside the JSON. The value of "label" must be exactly one of "CORRECT" or "INCORRECT". Use exactly this JSON structure: {"label": "CORRECT", "reason": "one short sentence explaining your judgment"}

CORRECT:

- The candidate correctly and sufficiently answers what the question asks.
- It does NOT need to reproduce the entire gold answer if the shorter answer fully answers the question.
- Harmless additional information is allowed.
- Different wording, word order, spelling variants, transliteration, punctuation, capitalization, or formatting are allowed when the meaning is unchanged.
- An equivalent answer in another language or script is allowed when it unambiguously refers to the same entity, value, date, place, or concept.
- Do not penalize Hebrew niqud (vowel diacritics) being present or absent.
- Do not penalize ktiv haser vs. ktiv male spelling differences (defective vs. full spelling).
- Do not penalize harmless grammatical or morphological differences, including prefixes such as the definite article.
- Equivalent date, number, unit, or title formatting is allowed.
- If multiple gold answers are provided, the candidate is CORRECT if it correctly matches any valid gold answer.

Examples:

Gold: "2002 בשנת" Candidate: "2002" → CORRECT

Gold: "24-ה" Candidate: "24 יום" → CORRECT

Gold: "ירושלים" Candidate: "ירושלים" → CORRECT

531 INCORRECT: 589
532 -- The core answer is factually wrong. 590
533 -- The candidate does not answer the
534 question.
535 -- The candidate contains a material
536 factual error or contradiction, even
537 if part of the answer is correct.
538 -- Merely being semantically related
539 to the gold answer is not sufficient.

540 Important rules:
541 -- Judge correctness, not lexical
542 overlap.
543 -- Judge whether the QUESTION has
544 been sufficiently answered, not
545 whether every word in the gold answer
546 was reproduced.
547 -- A shorter answer is CORRECT when
548 it contains all information necessary
549 to answer the question.
550 -- Extra information is allowed only
551 when it does not introduce a material
552 error.
553 -- Do not use F1, EM, model identity,
554 or experimental condition in your
555 judgment.

556 User message template: ``Question:
557 {question}\n\nGold answer(s):
558 {gold_answers}\n\nCandidate answer:
559 {prediction}``

560 A.2 Arabic judge

561 You are an Arabic QA evaluation
562 judge. Your task is to assess
563 whether a candidate answer correctly
564 answers a question, given the
565 reference (gold) answer(s).

566 Judge the candidate based on whether
567 it sufficiently and factually answers
568 the question relative to the provided
569 gold answer(s).

570 If a context passage is provided, use
571 it as additional evidence. If no
572 context is provided, judge based on
573 the question and gold answer(s).
574 Treat the provided gold answer(s) as
575 the reference; do not override them
576 using outside knowledge.

577 You must respond with valid JSON
578 only. No explanation outside the
579 JSON. The value of "label" must be
580 exactly one of "CORRECT" or
581 "INCORRECT". Use exactly this JSON
582 structure: {"label": "CORRECT",
583 "reason": "one short sentence
584 explaining your judgment"}

585 CORRECT:
586 -- The candidate correctly and
587 sufficiently answers what the
588 question asks.

-- It does NOT need to reproduce the
entire gold answer if the shorter
answer fully answers the question.

-- Harmless additional information is
allowed.

-- Different wording, word order,
spelling variants, transliteration,
punctuation, capitalization, or
formatting are allowed when the
meaning is unchanged.

-- An equivalent answer in another
language or script is allowed when it
unambiguously refers to the same
entity, value, date, place, or
concept.

-- Do not penalize Arabic diacritics
(Tashkeel) being present or absent.

-- Do not penalize harmless
grammatical/case-ending differences.

-- Equivalent date, number, unit, or
title formatting is allowed.

-- If multiple gold answers are
provided, the candidate is CORRECT if
it correctly matches any valid gold
answer.

Examples:

Gold: "2002 في عام" Candidate: "2002"
→ CORRECT

Gold: "1999م" Candidate: "1999" →
CORRECT

Gold: "مسقط" Candidate: "مَسْقَط" →
CORRECT

INCORRECT:

-- The core answer is factually wrong.

-- The candidate does not answer the
question.

-- The candidate contains a material
factual error or contradiction, even
if part of the answer is correct.

-- Merely being semantically related
to the gold answer is not sufficient.

Important rules:

-- Judge correctness, not lexical
overlap.

-- Judge whether the QUESTION has
been sufficiently answered, not
whether every word in the gold answer
was reproduced.

-- A shorter answer is CORRECT when
it contains all information necessary
to answer the question.

-- Extra information is allowed only
when it does not introduce a material
error.

-- Do not use F1, EM, model identity,
or experimental condition in your
judgment.

User message template: ``Question {question}\n\nGold answer(s): {gold\_answers}\n\nCandidate answer: {prediction}``

B Additional Diacritization Failure-Mode Patterns

Beyond the subject-echo pattern reported in Section 4.4, manual inspection of the ~900 judge-confirmed correct→incorrect pairs surfaces several further recurring patterns, mined primarily from entity-answer questions in HeQ and MKQA (Hebrew) where the gold answer is a short, checkable name. We report them here as candidates for the same logit-lens / patchscopes follow-up described in Section 5, since — like subject echo — each raises a distinct hypothesis about *where* in the network the correct entity is lost.

Entity-to-generic-noun blank. The correct proper noun is dropped in favor of a generic category noun, as though the model retained the semantic *type* of the answer (a person, a title) without the specific referent:

Q: *יש הכי הרבה מדליות זהב בעולם* ("Who has the most Olympic gold medals in the world?") — **Gold:** Michael Phelps — **Pred:** *למלך יש הרבה מדליות זהב בעולם* ("the king has many gold medals in the world")

Confusable-entity substitution (same category). The model names a real entity of the correct type and domain, but the wrong instance — swapping in another member of the same franchise, award category, or competition:

Q: "Who played Snape in the Harry Potter films?" — **Gold:** Alan Rickman — **Pred:** Rupert Grint (a different Harry Potter actor, playing Ron, not Snape)

Q: "Who won the Academy Award for Best Actress in 2018?" — **Gold:** Frances McDormand — **Pred:** Jennifer Lawrence (a different Best-Actress winner, wrong year)

Full/unrelated hallucination. The answer is a real, well-formed entity but has no discernible topical link to the question — a stronger failure than confusable substitution, since even the answer's category or domain is wrong:

Q: *איזו מדינה אירופית שלטה באחד דרום אפריקה* ("Which European country ruled the Union of South Africa?") — **Gold:** Britain — **Pred:** *אלג'יריה* ("Algeria" — not even a colonial power on that continent)

Category-break hallucination. The predicted answer switches *answer type* entirely, not just

referent — e.g. substituting an invented personal name where the question unambiguously calls for a country:

Q: *1958 בשנת במונדיאל בשנת* ("Who won the World Cup in 1958?") — **Gold:** Brazil — **Pred:** *פרנץ בורנשטיין* (an invented person's name, though the question asks which country won a team competition)

Numeric/format degradation. The correct magnitude is retrieved but rendered in a corrupted surface form — sign-flipped, unit-dropped, or otherwise malformed — suggesting the failure is localized to answer *formatting* rather than fact retrieval:

Q: *כמה פעמים בודקים בDoppelPaymer כל פניה?* ("How many times does DoppelPaymer check each request?") — **Gold:** 3 times — **Pred:** "−3" (correct magnitude, sign-flipped, unit word dropped)

Taken together with subject echo (Section 4.4), these six patterns span a spectrum from pure surface-format corruption (numeric/format degradation) through partial retrieval (entity-to-generic-noun blank, subject echo) to full semantic substitution (confusable and association-driven substitution) to total retrieval failure (full and category-break hallucination) — motivating layer-wise probing across this spectrum rather than treating diacritic-induced harm as a single failure type.